Exercise 1: A sequence of numbers is defined as:

$$a_1 = 2$$
$$a_2 = 5$$
$$a_n = a_{n-1} + a_{n-2}, \quad n \geq 3$$

Determine a_5 and discuss recursion tree.

Exercise 2: Give a recursive definition for each of the following sequences:

a) $a_n = 5n, \quad n \geq 1$

b) $a_n = 3n + 8, \quad n \geq 1$

c) $a_n = 2^n + 1, \quad n \geq 1$

d) $a_n = 6, \quad n \geq 1$

Exercise 3: Determine the value of the following summations:

a) $\sum_{k=2}^{5} k^2 + 2 + 3$

b) $\sum_{j=-2}^{3} (j^3 + 2) + 3$

Exercise 4: Express the following using the summation notation:

a) $1+2+3+4+5+6$

b) $5+6+7+8$

c) $2+2+2+2+2$

d) $\frac{2}{3!}+\frac{2}{4!}+\frac{2}{5!}+\cdots+\frac{2}{(n+2)!}$

Exercise 5: Given $\sum_{i=0}^{10}(2a_1 + 3a_2) = 209$ and $a_1 = 2$, find a_2.

Exercise 6: Given $\sum_{j=1}^{100} a_j = 500$ and $\sum_{k=0}^{49} a_{k+1} + \sum_{i=48}^{98} a_{i+2} + a_{50} = 506$, find a_{50}.

Exercise 7: Use formula $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$ to find:

a) Sum $10 + 11 + 12 + \ldots + n + (n+1)$ answer: $\frac{n^2 + 3n - 88}{2}$

b) The value of `counter` after the following programming segment is executed

```
counter = 10;
for (j = 6; j <= 100; j++)
        counter = counter + 2*(j + 4);
```

answer: `counter` = 10,840

c) The value of `time` after the following programming segment is executed

```
time = 20;
for (i = 1; i <= n; i++){
     time = time + 5
     for (j = 0; j <= n+2; j++){
               time = time + 10
     }
}
```

answer: `time` $= 10n^2 + 35n + 20$

d) The value of `time` after the following programming segment is executed

```
time = 20;
for (i = 1; i <= n; i++){
     time = time + 6i
     for (j = i; j <= n+2; j++){
               time = time + 10
     }
}
```

answer: `time` $= 8n^2 + 28n + 20$

e) The value of `time` after the following programming segment is executed

```
time = 20;
for (i = 1; i <= n; i++){
     time = time + 10i
     for (j = 4; j <= n+2; j++){
               time = time + 10j
     }
}
```

answer: `time` $= 5n^3 + 30n^2 - 25n + 20$

If all of the problems can't be done in the lab period, students can work on them on their own.